

Ultrapowers

If (X, d) is a metric space, we may consider the space $\ell^\infty(X)$ to be the space of all bounded sequences in X with metric defined by

$$d_\infty(x, y) = \sup_{n \in \mathbb{N}} d(x_n, y_n).$$

If ω is an ultrafilter on $\mathbb{N}$, then we define an equivalence relation $x \sim y$ on $\ell^\infty(X)$ by saying they are equal if

$$\lim_{\omega} d(x_n, y_n) = 0.$$

Then, we define the equivalence classes

$$[x]_{\omega} = \{y \in \ell^\infty(X) \mid y \sim x\}$$

and take the quotient $X_{\omega} = \ell^\infty(X)/\sim$, with distance metric defined by

$$d_{\omega}([x]_{\omega}, [y]_{\omega}) = \lim_{\omega} d(x_n, y_n).$$

This space is known as the *ultrapower* of the metric space (X, d) . The map $\iota_{\omega}: X \rightarrow X_{\omega}$ defined by $x \mapsto [(x, x, \dots)]_{\omega}$ is an isometric embedding of (X, d) into (X_{ω}, d_{ω}) .

Generally, we assume that the ultrafilter ω is nonprincipal, since if ω is principal, then taking the limit along the ultrafilter is equivalent to extracting the element of $\mathbb{N}$ that defines the principal ultrafilter.

A useful fact is that if (X, d) is complete, then so too is (X_{ω}, d_{ω}) .

Now, suppose G is a group, and let $d: G \times G \rightarrow [0, \infty)$ be a metric on G . We say the metric d is left (or right) invariant if $d(hg_1, hg_2) = d(g_1, g_2)$ for left invariance and $d(g_1h, g_2h) = d(g_1, g_2)$ for right invariance. A metric that is both left and right invariant is called bi-invariant, and the space (G, d) is called a metric group.

Example: Let G be a finitely generated group, and let $S \subseteq G$ be a finite generating subset. Then, (G, d_S) is a left metric group, where d_S denotes the word metric — i.e., $d_S(g, h)$ is the minimum word length necessary to construct the element gh^{-1} from S .

In the next section, we will discuss two important metric groups and various important properties thereof.

Sofic Groups

Suppose S_n is the group of permutations of the set $[n] = \{0, \dots, n-1\}$. Then, the *Hamming length* of a permutation $\sigma \in S_n$ is given by

$$\ell(\sigma) = \frac{1}{n} |\{i \in [n] : \sigma(i) \neq i\}|.$$

The metric $d: S_n \times S_n \rightarrow [0, \infty)$ given by $d(x, y) = \ell(xy^{-1})$ then is a bi-invariant length function.¹

Definition: A countable discrete group Γ is *sofic* if for every $\varepsilon > 0$ and every finite subset F of $\Gamma \setminus \{e\}$, there is n and a map $\Phi: \Gamma \rightarrow S_n$ such that $\Phi(e) = \text{id}$, and for every $g, h \in F \setminus \{e\}$, we have

- $d(\Phi(gh), \Phi(g)\Phi(h)) < \varepsilon$;
- $\ell(\Phi(g)) > r(g)$, where $r(g)$ is a constant depending only on g .

¹For invariance, recall that conjugation preserves cycle type, which is essentially what the Hamming distance is measuring. One can show that a metric being bi-invariant is equivalent to it being preserved under conjugation.

One of the main benefits of the ultrapower method is that it allows for a conversion from this local definition to a global definition. Fixing a free ultrafilter ω on $\mathbb{N}$, we define the ultrapower

$$\prod_{\omega} S_n := \prod_{n \in \mathbb{N}} S_n / \left\{ x \in \prod_{n \in \mathbb{N}} S_n : \lim_{\omega} \ell_{S_n}(x) = 0 \right\},$$

where $\ell_{S_n}(x) = d(x, \text{id})$.

It follows from this definition that Γ is sofic if and only if there exists an injective homomorphism $\Phi: \Gamma \rightarrow \prod_{\omega} S_n$ for any free ultrafilter ω on $\mathbb{N}$.

Hyperlinear Groups

References

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